

Micro I, class 1

Consider a nonempty set $X \subseteq \mathbb{R}_+^L$, a *consumption set*, describing all physically feasible consumption plans for a consumer. For price-wealth pair (p, w) with $p \in \mathbb{R}_{++}^L$ and $w \in \mathbb{R}_+$, the *budget set* is $B(p, w) = \{x \in X \mid p \cdot x \leq w\}$, the set of all financially feasible consumption plans. It is homogeneous of degree 0 in (p, w) : $B(\alpha p, \alpha w) = B(p, w)$ for every $p \gg 0$, $w \geq 0$, $\alpha > 0$.

Let $\succsim \subseteq X \times X$ be a binary relation describing consumer *preferences*. If $(x, y) \in \succsim$, we say that x is *at least as good as* y and usually write $x \succsim y$. Define $x \succ y$ (strict preference) by $x \succsim y$ but $y \not\succsim x$, and $x \sim y$ (indifference) by $x \succsim y$ and $y \succsim x$. The *demand correspondence* is

$$d(p, w) = \{y \in B(p, w) \mid y \succsim x \text{ for every } x \in B(p, w)\}$$

Even without making any assumptions on $\succsim$, it is clear that the demand correspondence is homogeneous of degree 0 in (p, w) : $d(\alpha p, \alpha w) = d(p, w)$ for every $p \gg 0$, $w \geq 0$, $\alpha > 0$, which is our first *testable implication* of consumer theory. It of course must also satisfy $d(p, w) \subseteq B(p, w)$.

We started a discussion of several axioms that are often imposed on consumer preferences: *completeness*, *transitivity*, *continuity* (sequential and mixture); *monotonicity* (regularly and strong, *local nonsatiation* (LNS)); and *convexity* (regular and strict). We'll finish this discussion on Tuesday. In ASMT I, you are exploring two special assumptions often made in applications: *homotheticity* and *quasilinearity*. The main event in the next class will be to give conditions under which a utility representation of preferences exists. This is closely related to **Proposition 3.C.1** in MWG.